Canvas
#01/10
Canvases:

PROBLEM SPACE
SOCIAL ROBOT CO-DESIGN CANVASES

Robot = AI System

What problem are you solving?

USER

Group(s)
Who are the users? Are there supporting users? For example: students and teachers.

primary users: ED triage nurses, Charge nurses

secondary users: Emergency physicians, patients, administrative staff

Characteristics
What are the users like?

Work under time pressure
High cognitive workload
Experienced clinicians

Depend on accurate triage
Mixed clinical knowledge
Need timely information

Needs
What do the users need?

Fast decision support
Explainable AI
Override capability
Simple interface

Accurate prioritisation
Efficient patient flow
Safe communication

Goal(s)
What do the primary and secondary users want to accomplish?

SHORT-TERM

primary users:
Complete rapid triage
Prioritise high-risk patients

secondary users:
Receive accurate patient information
Improve coordination

LONG-TERM

primary users:
Reduce fatigue
Improve consistency

secondary users:
Improve patient outcomes
Reduce waiting times

ROBOT

Task(s)
What task does the robot perform?

SHORT-TERM

Analyse patient triage information
Predict ESI level
Explain recommendation
Support nurse decision-making

LONG-TERM

Improve triage consistency
Reduce decision fatigue
Support continuous model improvement through override feedback

Advantages
What advantages does using a robot bring (compared to a computer or human)?

General Advantages

Social skills

User's emotional response

Personalization

Precise tasks

Consistent triage recommendations
Explainable AI improves clinician trust
Rapid analysis of multiple clinical variables
Real-time decision support
Seamless integration with the EHR and hospital workflow

Data collection with sensors

Mobility

Environment manipulation

Electronic Health Record (EHR)
Vital sign monitoring devices
Hospital patient database
Staff notification system
Secure audit log for overrides
and quality improvement

Connection to systems

Canvas #02/10

Canvases:

ETHICAL CONSIDERATIONS

SOCIAL ROBOT CO-DESIGN CANVASES

Robot = AI System

Consider potential ethical problems, and potential solutions –both from the user's and robot's perspectives.
Consider the boxes to be guidelines: you don't need to fill each one.

Physical safety

Machines can pinch or crush the user. How is this mitigated?

	PROBLEM	SOLUTION
USER	Incorrect recommendation	Nurse makes final decision
ROBOT	Poor-quality input data	Request manual verification

Data security

Is the robot in a unique data collection position? How is the user's data protected?

	PROBLEM	SOLUTION
USER	Patient privacy	Encrypted access & privacy filters
ROBOT	Secure storage	Audit logs & encryption

Transparency

How does the robot share an accurate perception of its abilities, intentions and constraints, so the user can evaluate their trust in it?

	PROBLEM	SOLUTION
USER	Recommendation unclear	Show confidence + top factors
ROBOT	Seen as replacing clinicians	Label as AI recommendation

Equality across users

Robots' algorithms can be biased. A robot's appearance could reinforce harmful stereotypes. What are potential issues?

	PROBLEM	SOLUTION
USER	Possible demographic bias	Monitor performance & retrain
ROBOT	Dataset limitations	Regular bias audits

Emotional consideration

People have been shown to form emotional attachments to robots, as if they were alive. Is this a potential problem?

	PROBLEM	SOLUTION
USER	Over-trust or distrust	Explain predictions clearly
ROBOT	False confidence	Display confidence score and explain reasoning

Behaviour enforcement

People could transfer their inappropriate behaviour, such as rudeness, from robots to humans. How is this mitigated?

	PROBLEM	SOLUTION
USER	Ignoring guidance	Training + override logging
ROBOT	Unsafe use	Maintain audit trail

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#03/10

Canvases:

DESIGN GUIDELINES

SOCIAL ROBOT CO-DESIGN CANVASES

What things are important to consider in the robot's design?

Advantage guidelines

What advantages can the robotic solution have?
Think back to what you defined in the solution space canvas.

What advantages should the AI system provide?

Rapid triage decision support
Explain recommendations
Consistent ESI classification
Integrates into existing workflow
Supports, not replaces, clinicians

Ethical guidelines

What ethical considerations does the robot have?
Think back to what you defined in the ethics canvas.

Protect patient privacy
Maintain clinician oversight
Minimise algorithmic bias
Record override decisions
Display confidence and explanations

ROBOT DIMENSIONS

Robot = AI System
Form = Interface

Environment guidelines

What should the robot's context be like?
For example:
- If users are especially vulnerable, should it optimize for support?
- If the robot is part of a strict process, should it optimize for efficiency and security?

Optimised for busy EDs
Works during day and night shifts
High readability under stress
Minimal cognitive load
Resilient to network interruptions

Form guidelines

What guides the design of the robot's outward qualities?
For example:
- Should the robot be designed to appear especially approachable, or more industrial?
- Should it be simple, or detailed?

Clean clinical dashboard
Simple information hierarchy
High-contrast colours
Large readable text
Minimal screen clutter

Interaction guidelines

What guides the design of interaction?
For example:
- Is the interaction multimodal, or is one modality optimized for efficiency?
- Should the user feel empowered and lead the interaction, or does the robot provide safety via leadership?
- Is the goal of the interaction to complete a task, or explore?

Touchscreen and mouse/keyboard
Human leads all decisions
AI provides recommendations
Explainability shown before action
Fast, efficient workflow

Behaviour guidelines

What guides the design of the robot's behaviour?
For example:
- Should behaviour be simple, or sensitive to context?
- Does the robot have internal drivers, or does it react to external stimuli?
- Does the robot have social skills?

Responds in real time
Flags missing data
Requests updated vital signs
Escalates high-risk patients
Adapts to changing patient information

THRI paper:

Canvas #04/10

Canvases:

ROBOT DESIGN MVP

SOCIAL ROBOT CO-DESIGN CANVASES

Robot = AI System

It's time to design your robot MVP (Minimum Viable Product)! Remember the guidelines you defined.

Where and when

What place? Mercer General ED triage desk

What time of day? 24/7 operation

Does the place or time change? Optimised for peak workload

Robot's role

Is the robot a friend? Teacher? Helper? Something else?

Clinical decision-support assistant, Explainable AI advisor, Workflow support tool

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

Personality

Does the robot have specific characteristics?
Does it have emotional states, or needs?

Calm, professional, transparent, non-judgemental, supportive

Context-based behaviour

What external and environmental factors affect behaviour?
What data is used to adapt to context?

Updates recommendations as new data arrive, Flags incomplete information, Highlights deteriorating patients, Displays hospital capacity alerts, Supports mass casualty workflows

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

EHR, vital signs monitors, hospital database, staff notification system

Interaction modalities

What modalities are inputs to the robot? What modalities does the robot output?

INPUT

- [ ] voice
- [ ] sounds
- [ ] gestures
- [ ] movement
- [x] touch
- [ ] smell
- [ ] facial expressions
- [x] screens / buttons
- [ ] lights
- [ ] other text, system connections (EHR)

External sensors (vital signs)

OUTPUT

- [ ] voice
- [x] sounds alarms/
- [ ] gestures
- [ ] movement
- [ ] touch
- [ ] smell
- [ ] facial expressions
- [x] screens
- [ ] lights
- [ ] other text

system connections, visual alerts

Interaction flow

Describe the most important interaction of the robot.
Note: only fill the bottom row if your robot is teleoperated.

General Interaction Flow

	BEFORE	DURING	AFTER
USER	1. Nurse enters/reviews patient information.		4. Nurse accepts or overrides the recommendation.
ROBOT	2. AI analyses available clinical data.	3. Predicted ESI level, confidence score and contributing factors are displayed.	5. Decision is recorded in the EHR and audit log.
ROBOT OPERATOR (optional)			

Cite as: Axelsson, M., Oliveira, R., Racca, M., & Kyrki, V. (2021). Social robot co-design canvases: A participatory design framework. ACM Transactions on Human-Robot Interaction (THRI), 11(1), 1-39.

THRI paper:

Canvas
#05/10
Canvases:

ENVIRONMENT
SOCIAL ROBOT CO-DESIGN CANVASES

Robot = AI System

What is the robot's context of operation?
You can use the "Ecosystem" canvas to dive deeper into this topic.

Where
What place?
Does it change?

Mercer General ED
Main triage desk

User(s)
Who is using the robot?

Triage nurses

When
What time of day?
Does it change?

24/7
Peak and night shifts

Secondary user(s)
Are there secondary users?
E.g. teachers that help students use a robot.

Physicians
Nurse supervisors
Administrative staff

Data collection
Does the robot collect data from its environment?
How is it stored?

Electronic Health Record (EHR)
Manual entry
Vital signs
Secure hospital database

TRADE-OFF:
More data collection requires more attention to data security.

Simultaneous users
How many users should be able to use the robot simultaneously?

TRADE-OFF:
More simultaenous users requires a more sophisticated robot.

one — few — several

One patient per workstation

External sensors and actuators
Does the robot use external sensors?
Does it have external actuators, such as lights or limbs?

Vital signs monitors
No physical actuators

Connection to systems
Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

EHR
Vital signs devices
Hospital network
Audit database

Canvas
#06/10
Canvases:

Robot = AI System

Form = Interface

FORM

SOCIAL ROBOT CO-DESIGN CANVASES

What are the robot's outward qualities? If an existing robot is used, are its qualities modified?

Draw a picture

What does the robot look like?
Is it attached to something?
Does it move around?
Can its appearance be modified?

See Interface Mock-Up Design

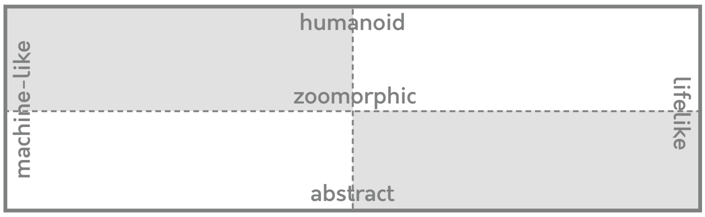
Appearance

Clean clinical dashboard
High-contrast colours
Large readable typography
Simple information hierarchy
Minimal visual clutter while still maintaining everything necessary

Appearance

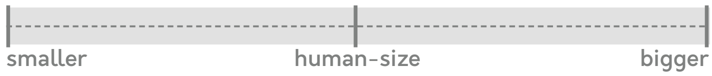
Is the robot more machine or lifelike? Is it human-shaped, animal-shaped, or abstract?

TRADE-OFF:
Robots that appear more human and lifelike are expected to be more sophisticated in features.



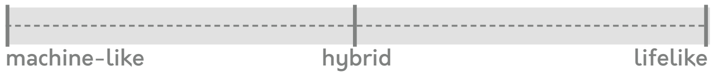
Size

How big is the robot?



Character of movement

What is the robot's movement like?



Voice & sounds

Does the voice have a gender or an age? What are pitch, speed and prosody like? Is the voice always the same?
Does the robot make sounds: music, "beep"s, animal noises?
When are these sounds heard?

Audio & Notifications

Visual-first alerts
Optional notification sounds for critical events
No continuous alarms to reduce alarm fatigue

Mobility

Does the robot move across space? Does it move in place?

Navigation & Layout

Fixed desktop dashboard
Card-based layout
Frequently used actions prominently displayed
Information organised for rapid scanning

Visual cues

Does the robot have expressions, lights, a screen or other visual elements?

Colour-coded ESI levels for first glances
Confidence indicator
Explainability panel
Missing information warnings
Capacity alerts
Queue summary
Timestamp indicators

Touch & smell sensations

Is the robot soft or rough, warm or cold?
How does the robot smell?
Touch and smell are especially important in close interactions.

Accessibility

Large touch targets
Keyboard and touchscreen compatible
WCAG-inspired colour contrast
Clear labels and icons
Designed for users under cognitive stress

Canvas #07/10

Canvases:

INTERACTION

SOCIAL ROBOT CO-DESIGN CANVASES

Robot = AI System

How does the robot interact with the user?
You can use the “Experience Flow” canvas to dive deeper into this topic.

Interaction modalities

What modalities are inputs to the robot? What modalities does the robot output?

INPUT

- [ ] voice
- [ ] sounds
- [ ] gestures
- [ ] movement
- [x] touch
- [ ] smell
- [ ] facial expressions
- [x] screens
- [ ] lights
- [ ] other text, system (EHR) connections

External sensors (vital signs)

OUTPUT

- [ ] voice
- [x] alarms/ sounds
- [ ] gestures
- [ ] movement
- [ ] touch
- [ ] smell
- [ ] facial expressions
- [x] screens
- [ ] lights
- [ ] other text

system connections, visual alerts

Interaction flow

Describe the most important interaction of the robot.
Note: only fill the bottom row if your robot is teleoperated.

	BEFORE	DURING	AFTER
USER	1. Nurse enters/reviews patient information.		4. Nurse accepts or overrides the recommendation.
ROBOT	2. AI analyses available clinical data.	3. Predicted ESI level, confidence score and contributing factors are displayed.	5. Decision is recorded in the EHR and audit log.
ROBOT OPERATOR (optional)			

Situation flow

How defined is the situation where the interaction takes place? Does the user always enter and exit at the same point?

predefined — flexible — freestyle

Leadership

Who initiates the interaction? Who determines what happens next?

robot-led — mutual / alternate — user-led

Goal

What is the user’s goal in the interaction? What describes the interaction?

task completion / informative — both — explorative

Robot’s name

Does the robot have a name which is used during interaction?

TRADE-OFF:
A robot with a name, creates more emotional bond.

Canvas
#08/10

Canvases:

BEHAVIOUR

Robot = AI System

SOCIAL ROBOT CO-DESIGN CANVASES

What factors guide the robot's behaviour?

Robot's role

Is the robot a friend? Teacher? Helper? Something else?

Explainable clinical decision-support assistant
Supports ED triage workflow
Improves consistency and patient safety

Motivation

How is the robot's behaviour motivated? Is it based on external data, internal models such as personality, or both?

external / environment based — both — internal

External clinical data (patient demographics, chief complaint, vital signs and EHR data) and the trained Random Forest Mode. No emotional/personality-based behaviour is used.

Personality

Does the robot have specific characteristics? Does it have emotional states, or needs?

Interaction Style

Calm
Professional
Transparent
Reliable
Non-intrusive

TRADE-OFF:
More personality creates more emotional bond.

Mode of operation

Is the robot operating by itself, or is a human affecting behaviour? Is a human in full control?

TRADE-OFF:
A human-operated robot requires a good user interface, an autonomous robot requires a good control logic.

fully human-operated — partial human control — autonomous

AI generates recommendations
Nurse reviews, accepts or overrides
Final clinical decision always made by the nurse

Social skills

How good are the robot's social skills: does it greet a new person and ask their name? Does it follow people with its gaze?

TRADE-OFF:
Extensive social skills require a more sophisticated robot.

none — some — extensive

Social behaviours

What social behaviours does the robot exhibit?

Clear, respectful communication through simple language and visual explanations. No conversational or human-like social behaviours.

Contextual adaptation

Does the robot's behaviour vary according to context, e.g. by weather or time of day?

TRADE-OFF:
More contextual adaptation requires a more sophisticated robot.

none — some — extensive

Adapts to patient vital signs, responds to changing clinical information, displays hospital capacity alerts, supports mass casualty mode when activated

Context-based behaviour

What external and environmental factors affect behaviour? What data is used to adapt to context?

Updated vital signs
New clinical information
Missing data
ED capacity
Mass casualty incidents
Low-confidence predictions

Personalization

Does the robot behave differently toward different people? Does it need to remember people, and store their data?

No personalisation of behaviour
Recommendations are based only on the patient's clinical data
No long-term user profiling
Stores information only for clinical documentation and audit purposes

TRADE-OFF:
More personalization requires more personal data from the user.

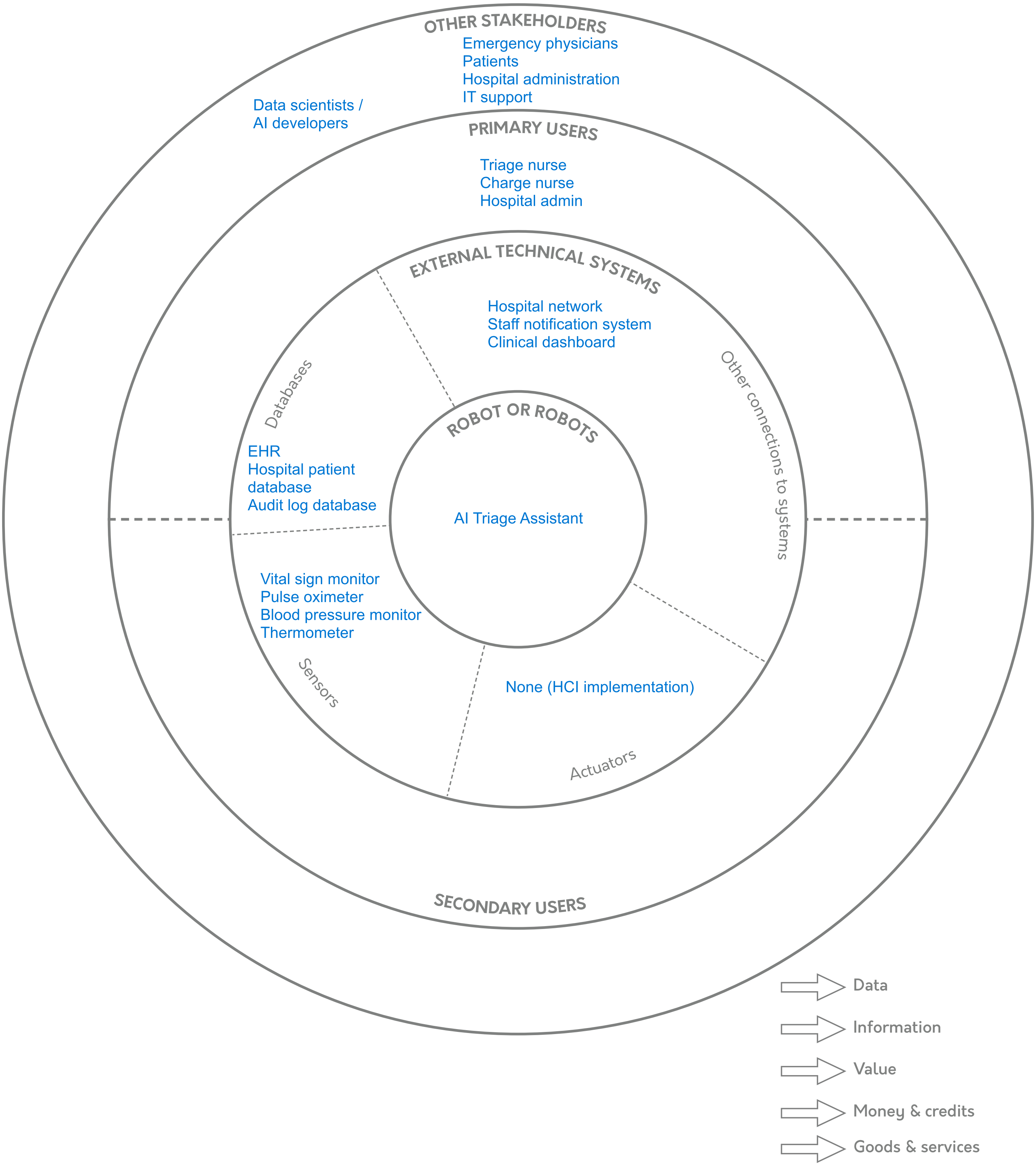
Canvas #09/10

Canvases:

SERVICE ECOSYSTEM

SOCIAL ROBOT CO-DESIGN CANVASES

What stakeholders does the robot's operation involve?
Draw sectors for different stakeholders.
Select a color for each of the resources, and draw arrows to show their flow from stakeholder to stakeholder.



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Canvas #10/10

Canvases:

EXPERIENCE FLOW

SOCIAL ROBOT CO-DESIGN CANVASES

Robot = AI System

Describe the most important interaction of the robot.
Note: only fill the bottom row if your robot is teleoperated.

		BEFORE	DURING	AFTER
USER	FEELING e.g. confused	Focused, preparing	Busy, under pressure	Confident, informed
USER	THINKING e.g. “I need help.”	"What information do I have?"	"Does this recommendation make sense?"	"Have I prioritised correctly?"
USER	DOING e.g. pushes button	Enter/review patient details	Review AI recommendation, accept or override	Complete triage and document decision
ROBOT	DOING e.g. says “Hello!”	Waits for patient data	Analyses data, predicts ESI, explains recommendation	Stores prediction and audit record
ROBOT	SENSOR INPUT e.g. sees user’s face	Collect vital signs	Detect updates or missing data	Save latest observations
ROBOT	CONNECTION TO SYSTEMS e.g. records data in database	Retrieve patient record	Query EHR, update dashboard	Save final triage outcome and override reasoning
ROBOT OPERATOR (optional)	DOING e.g. controls robot’s arm	Prepare for patient assessment	Validate AI recommendation and intervene if necessary	Finalise ESI level and initiate appropriate care

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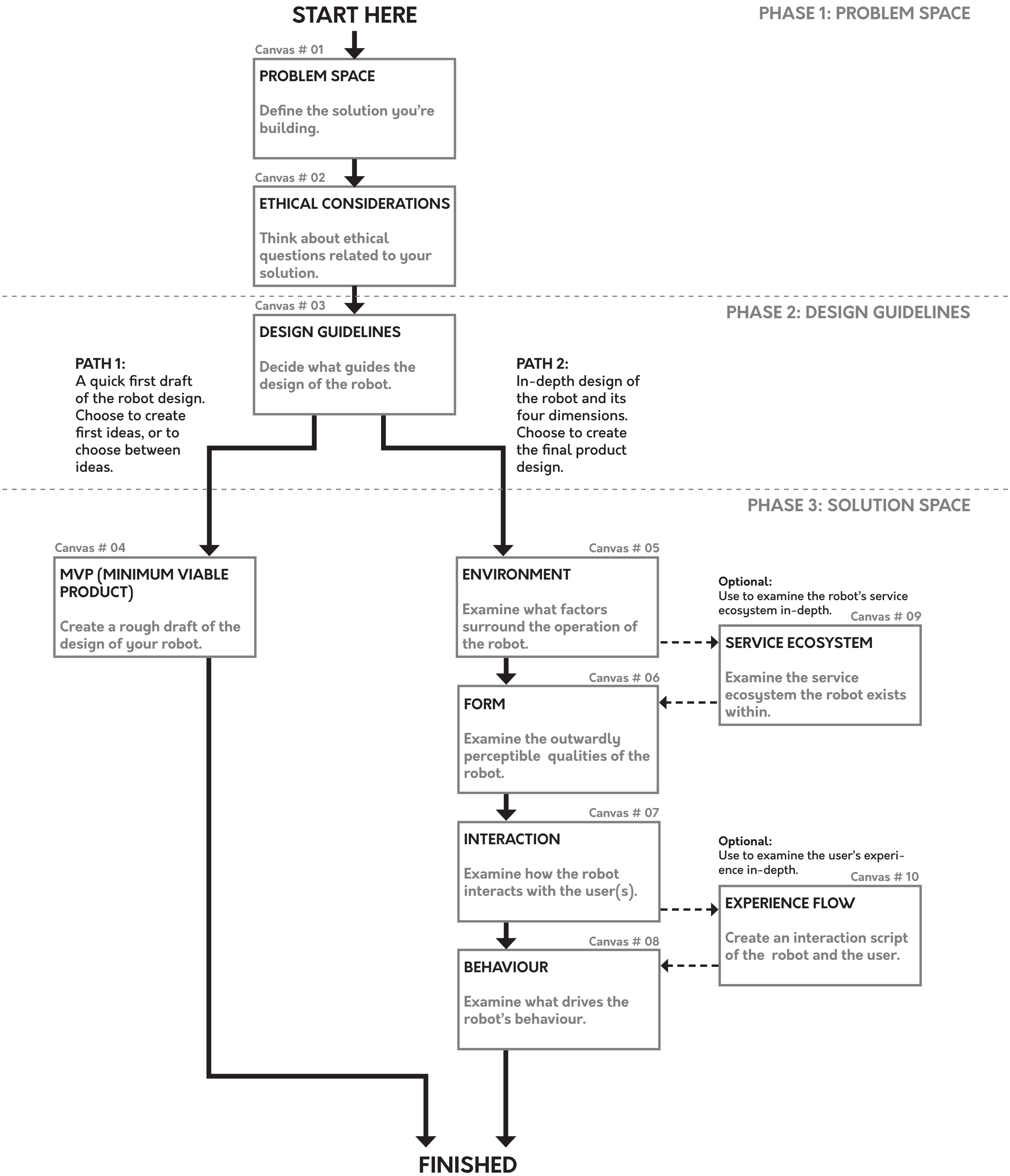
DESIGN PATH

SOCIAL ROBOT CO-DESIGN CANVASES

How to choose your canvases

Canvases:



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